Grade 10, Term 2, Lesson 1, Criterion C1

Outcomes, Support and Probability Mass Functions

C1. Characterises discrete random variables in economic and financial contexts by identifying their support, probability mass function and possible outcomes.

All data are hypothetical teaching models. Keep exact expressions where useful and round final probabilities to four decimal places.

Highlight legend

Highlighted item	What it means	Common symbols
Outcome	A possible result of the experiment.	ω (one outcome), Ω (sample space).
Probability	Probability expressions and PMF masses, without units.	$P(\omega)$, $p_X(x)$; e.g. result: 0.4.
First variable	The first variable, its numerical values, and any units.	F, X; values f, x.
Second variable	The second variable, its numerical values, and any units.	E, Y; values e, y.
Third variable	The third variable, its numerical values, and any units.	C, Z; values c, z.
Unknown parameter	A quantity to determine from the information given.	k (illustrative only): this deck uses no unknown parameter.

The problem and your solution

Given complete, mutually exclusive and exhaustive outcomes, their probabilities, and a rule assigning numerical values:

Task 1. List outcomes and probabilities.

Task 2. Define the variable and units; map outcomes to values; give the support and PMF; check validity.

Task 3. Calculate a requested probability and interpret it in context.

$$\text{Outcomes} \rightarrow \begin{matrix}\text{definition}\\ \text{and mapping}\end{matrix} \rightarrow \begin{matrix}\text{support}\\ \text{and PMF}\end{matrix} \rightarrow \text{checks} \rightarrow \begin{matrix}\text{requested probability}\\ \text{and interpretation}\end{matrix}$$

General model: Mapping, PMF and checks

Outcome and mapping table

One row per complete outcome.

Task 1	
ω_i	$q_i = P(\{\omega_i\})$
ω_1	q_1
ω_2	q_2
$\vdots$	$\vdots$
ω_n	q_n

General model: Mapping, PMF and checks

Task 2: X is [quantity measured]; units: [units or unitless].

Outcome and mapping table

One row per complete outcome.

Task 1		Task 2
ω_i	$q_i = P(\{\omega_i\})$	$X(\omega_i)$
ω_1	q_1	$X(\omega_1)$
ω_2	q_2	$X(\omega_2)$
$\vdots$	$\vdots$	$\vdots$
ω_n	q_n	$X(\omega_n)$

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$\vdots$	$\vdots$	$\vdots$
ω_n	q_n	$X(\omega_n)$

Task 2: PMF (group equal values)
One row per distinct value.

x_j	$p_X(x_j) = \sum_{i:\, X(\omega_i)=x_j} q_i$
x_1	$\sum_{i:\, X(\omega_i)=x_1} q_i$
$\vdots$	$\vdots$
x_m	$\sum_{i:\, X(\omega_i)=x_m} q_i$

Task 2 — Support: $S_X = \{x : p_X(x) > 0\}$; units: [same as X].

General model: Mapping, PMF and checks

Task 2: X is [quantity measured]; units: [units or unitless].

Outcome and mapping table
One row per complete outcome.

	Task 1	Task 2
ω_i	$q_i = P(\{\omega_i\})$	$X(\omega_i)$
ω_1	q_1	$X(\omega_1)$
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$\vdots$	$\vdots$
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Task 2 — Support: $S_X = \{x : p_X(x) > 0\}$; units: [same as X].

Task 2 — Checks: $p_X(x) \geq 0$ for every x, and $\sum_{x \in S_X} p_X(x) = 1$.

General model: Mapping, PMF and checks

Task 2: X is [quantity measured]; units: [units or unitless].

Outcome and mapping table
One row per complete outcome.

	Task 1	Task 2
ω_i	$q_i = P(\{\omega_i\})$	$X(\omega_i)$
ω_1	q_1	$X(\omega_1)$
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$\vdots$	$\vdots$	$\vdots$
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Task 2 — Support: $S_X = \{x : p_X(x) > 0\}$; units: [same as X].

Task 2 — Checks: $p_X(x) \geq 0$ for every x, and $\sum_{x \in S_X} p_X(x) = 1$.

Task 3: A = [requested values]; $P(X \in A) = \sum_{x \in S_X \cap A} p_X(x)$.

General model: Mapping, PMF and checks

Task 2: X is [quantity measured]; units: [units or unitless].

Outcome and mapping table
One row per complete outcome.

	Task 1	Task 2
ω_i	$q_i = P(\{\omega_i\})$	$X(\omega_i)$
ω_1	q_1	$X(\omega_1)$
ω_2	q_2	$X(\omega_2)$
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Task 3: A = [requested values]; $P(X \in A) = \sum_{x \in S_X \cap A} p_X(x)$.

Task 3 — Interpret: [The chance that [quantity] is in [requested set] is [result].]

Payment fee: Mapping, PMF and checks

A school shop wants to understand the payment fee associated with its next sale. Customers use different payment methods, with different fees. Before the next payment arrives, what fees are possible, and how likely is each one?

Payment fee: Mapping, PMF and checks

Outcome and mapping table
One row per complete outcome.

Task 1	
ω	$P(\omega)$
Cash	0.1
Debit card	0.2
Credit card	0.3
Mobile wallet	0.4

Payment fee: Mapping, PMF and checks

Task 2: F is the payment fee in USD.

Outcome and mapping table

One row per complete outcome.

Task 1		Task 2
ω	$P(\omega)$	$F(\omega)$
Cash	0.1	0
Debit card	0.2	1
Credit card	0.3	1
Mobile wallet	0.4	2

Predict. Which methods share a fee?

Payment fee: Mapping, PMF and checks

Task 2: F is the payment fee in USD.

Outcome and mapping table
One row per complete outcome.

	Task 1	Task 2
ω	$P(\omega)$	$F(\omega)$
Cash	0.1	0
Debit card	0.2	1
Credit card	0.3	1
Mobile wallet	0.4	2

Task 2: PMF (group equal fees)
Predict. Which methods share a fee?

f (USD)	$p_F(f) = P(F = f)$
0	$p_F(0) = 0.1$
1	$p_F(1) = 0.2 + 0.3 = 0.5$
2	$p_F(2) = 0.4$

Task 2 — Support: $S_F = \{0, 1, 2\}$; values are measured in USD.

Payment fee: Mapping, PMF and checks

Task 2: F is the payment fee in USD.

Outcome and mapping table
One row per complete outcome.

	Task 1	Task 2
ω	$P(\omega)$	$F(\omega)$
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Debit card	0.2	1
Credit card	0.3	1
Mobile wallet	0.4	2

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Task 2 — Support: $S_F = \{0, 1, 2\}$; values are measured in USD.
Task 2 — Checks: All masses are nonnegative; $0.1 + 0.5 + 0.4 = 1$.

Payment fee: Mapping, PMF and checks

Task 2: F is the payment fee in USD.

Outcome and mapping table
One row per complete outcome.

	Task 1	Task 2
ω	$P(\omega)$	$F(\omega)$
Cash	0.1	0
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Task 2 — Support: $S_F = \{0, 1, 2\}$; values are measured in USD.
Task 2 — Checks: All masses are nonnegative; $0.1 + 0.5 + 0.4 = 1$.
Task 3: What is the probability of a 1 USD fee?

Payment fee: Mapping, PMF and checks

Task 2: F is the payment fee in USD.

Outcome and mapping table
One row per complete outcome.

	Task 1	Task 2
ω	$P(\omega)$	$F(\omega)$
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Task 2: PMF (group equal fees)
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Task 2 — Support: $S_F = \{0, 1, 2\}$; values are measured in USD.
Task 2 — Checks: All masses are nonnegative; $0.1 + 0.5 + 0.4 = 1$.
Task 3: What is the probability of a 1 USD fee? $P(F = 1) = 0.5$.

Payment fee: Mapping, PMF and checks

Task 2: F is the payment fee in USD.

Outcome and mapping table
One row per complete outcome.

	Task 1	Task 2
ω	$P(\omega)$	$F(\omega)$
Cash	0.1	0
Debit card	0.2	1
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Task 2: PMF (group equal fees)
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Task 2 — Support: $S_F = \{0, 1, 2\}$; values are measured in USD.
Task 2 — Checks: All masses are nonnegative; $0.1 + 0.5 + 0.4 = 1$.
Task 3: What is the probability of a 1 USD fee? $P(F = 1) = 0.5$.
Interpret: The next payment has a 50% chance of carrying a 1 USD fee.

Payment outcomes and their assigned values

A school shop receives one payment. The payment method is the *complete outcome*. A *random variable* assigns a number to each outcome.

Outcome ω	$P(\omega)$
Cash	0.1
Debit card	0.2
Credit card	0.3
Mobile wallet	0.4

Payment outcomes and their assigned values

A school shop receives one payment. The payment method is the *complete outcome*. A *random variable* assigns a number to each outcome.

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)
Cash	0.1	0
Debit card	0.2	1
Credit card	0.3	1
Mobile wallet	0.4	2

F is the payment fee in USD.

Payment outcomes and their assigned values

A school shop receives one payment. The payment method is the *complete outcome*. A *random variable* assigns a number to each outcome.

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$
Cash	0.1	0	0
Debit card	0.2	1	1
Credit card	0.3	1	1
Mobile wallet	0.4	2	1

F is the payment fee in USD.
E is 1 for an electronic payment and 0 otherwise (unitless).

Payment outcomes and their assigned values

A school shop receives one payment. The payment method is the *complete outcome*. A *random variable* assigns a number to each outcome.

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

F is the payment fee in USD.
E is 1 for an electronic payment and 0 otherwise (unitless).
C is 1 for a card payment and 0 otherwise (unitless).

Supports of the fee and indicators

The *support* is the set of values with positive probability.

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

F is the payment fee in USD.
E is 1 for an electronic payment and 0 otherwise (unitless).
C is 1 for a card payment and 0 otherwise (unitless).

Predict. Which distinct values appear in each variable's column?

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The *support* is the set of values with positive probability.

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

F is the payment fee in USD.

E is 1 for an electronic payment and 0 otherwise (unitless).

C is 1 for a card payment and 0 otherwise (unitless).

Predict. Which distinct values appear in each variable's column?

F : $\{0, 1, 2\}$ (USD) E : $\{0, 1\}$ (unitless) C : $\{0, 1\}$ (unitless)

Probability of a 1 USD fee

A *probability mass function* (PMF) gives each value's probability.

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

F is the payment fee in USD.

Predict. Which outcomes give $F = 1$?

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Cash	0.1	0	0	0
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Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

F is the payment fee in USD.

Predict. Which outcomes give $F = 1$?

Debit card and Credit card are mutually exclusive, so add their probabilities.

Probability of a 1 USD fee

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Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

F is the payment fee in USD.

Predict. Which outcomes give $F = 1$?

Debit card and Credit card are mutually exclusive, so add their probabilities.

$$P(F = 1) = 0.2 + 0.3 = 0.5.$$

Probability of an electronic payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

E is 1 for an electronic payment and 0 otherwise (unitless).

Predict. Which outcomes give $E = 1$?

Probability of an electronic payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

E is 1 for an electronic payment and 0 otherwise (unitless).

Predict. Which outcomes give $E = 1$?
Debit card, Credit card and Mobile wallet.

Probability of an electronic payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

E is 1 for an electronic payment and 0 otherwise (unitless).

Predict. Which outcomes give $E = 1$?
Debit card, Credit card and Mobile wallet.

$$P(E = 1) = 0.2 + 0.3 + 0.4 = 0.9.$$

Probability of a card payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

C is 1 for a card payment and 0 otherwise (unitless).

Predict. Which outcomes give $C = 1$?

Probability of a card payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

C is 1 for a card payment and 0 otherwise (unitless).

Predict. Which outcomes give $C = 1$?
Debit card and Credit card.

Probability of a card payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

C is 1 for a card payment and 0 otherwise (unitless).

Predict. Which outcomes give $C = 1$?
Debit card and Credit card.

$$P(C = 1) = 0.2 + 0.3 = 0.5.$$

The fee PMF

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

Fee value f (USD)	0	1	2
$P(F = f)$	0.1	0.5	0.4

Worksheet reading. See the worked grouping problem for the full fee calculation.

The electronic-payment and card-payment PMFs

E is 1 for an electronic payment and 0 otherwise (unitless).
C is 1 for a card payment and 0 otherwise (unitless).

Electronic-payment value e	0	1
$P(E = e)$	0.1	0.9

Card-payment value c	0	1
$P(C = c)$	0.5	0.5

E and C share a support but have different masses. All probabilities are unitless.

Three independent visitor decisions

A school enterprise offers a 5 USD notebook to each of three visitors.

Decision	Symbol	Probability
Buy	B	0.4
Pass	N	0.6

Assume the visitors decide *independently*. One visitor's decision leaves the other visitors' probabilities the same.

Eight ordered outcomes

Choose B or N for each visitor. Read the three positions in visitor order.

Predict. List the complete outcomes with zero, one, two and three buyers.

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$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}$$

Eight ordered outcomes

Choose B or N for each visitor. Read the three positions in visitor order.

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$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}$$

Eight ordered outcomes

Choose B or N for each visitor. Read the three positions in visitor order.

Predict. List the complete outcomes with zero, one, two and three buyers.

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Probability that only the first visitor buys

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

In BNN, only the first visitor buys. Independence lets us multiply the three decision probabilities in visitor order.

Predict. Which factor belongs to each position in BNN?

$$P(\mathrm{BNN})$$

Probability that only the first visitor buys

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

In BNN, only the first visitor buys. Independence lets us multiply the three decision probabilities in visitor order.

Predict. Which factor belongs to each position in BNN?

$$P(\mathrm{BNN}) = 0.4$$

Probability that only the first visitor buys

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

In BNN, only the first visitor buys. Independence lets us multiply the three decision probabilities in visitor order.

Predict. Which factor belongs to each position in BNN?

$$P(\mathrm{BNN}) = 0.4 \times 0.6$$

Probability that only the first visitor buys

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

In BNN, only the first visitor buys. Independence lets us multiply the three decision probabilities in visitor order.

Predict. Which factor belongs to each position in BNN?

$$P(\mathrm{BNN}) = 0.4 \times 0.6 \times 0.6$$

Probability that only the first visitor buys

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

In BNN, only the first visitor buys. Independence lets us multiply the three decision probabilities in visitor order.

Predict. Which factor belongs to each position in BNN?

$$P(\mathrm{BNN}) = 0.4 \times 0.6 \times 0.6 = 0.144.$$

Outcomes with the same probability product

$$\Omega = \{\text{NNN}, \text{BNN}, \text{NBN}, \text{NNB}, \text{BBN}, \text{BNB}, \text{NBB}, \text{BBB}\}.$$

Predict. Which outcomes have the same factors, possibly in a different order?

Complete outcomes	Probability of each outcome

Outcomes with the same probability product

$$\Omega = \{\text{NNN}, \text{BNN}, \text{NBN}, \text{NNB}, \text{BBN}, \text{BNB}, \text{NBB}, \text{BBB}\}.$$

Predict. Which outcomes have the same factors, possibly in a different order?

Complete outcomes	Probability of each outcome
NNN	
BNN, NBN, NNB	
BBN, BNB, NBB	
BBB	

Outcomes with the same probability product

$$\Omega = \{\text{NNN}, \text{BNN}, \text{NBN}, \text{NNB}, \text{BBN}, \text{BNB}, \text{NBB}, \text{BBB}\}.$$

Predict. Which outcomes have the same factors, possibly in a different order?

Complete outcomes	Probability of each outcome
NNN	$0.6^3 = 0.216$
BNN, NBN, NNB	$0.4 \times 0.6 \times 0.6 = 0.144$
BBN, BNB, NBB	$0.4^2 \times 0.6 = 0.096$
BBB	$0.4^3 = 0.064$

Buyer count, revenue and first visitor

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Variable	Finite support	Units
X counts the buyers among the three visitors.	$\{0, 1, 2, 3\}$	buyers
$Y = 5X$ is total notebook revenue.	$\{0, 5, 10, 15\}$	USD
Z is 1 if the first visitor buys and 0 if the first visitor passes.	$\{0, 1\}$	Unitless

Comparing BNN and NBN

In visitor order, B means buy and N means pass.
X counts the buyers among the three visitors.
$Y = 5X$ is total notebook revenue in USD.
Z is 1 if the first visitor buys and 0 if the first visitor passes (unitless).

Predict. Which values agree for these two outcomes?

Outcome	X (buyers)	Y (USD)	Z (unitless)
BNN			
NBN			

Comparing BNN and NBN

In visitor order, B means buy and N means pass.
X counts the buyers among the three visitors.
$Y = 5X$ is total notebook revenue in USD.
Z is 1 if the first visitor buys and 0 if the first visitor passes (unitless).

Predict. Which values agree for these two outcomes?

Outcome	X (buyers)	Y (USD)	Z (unitless)
BNN	1	5	
NBN	1	5	

Both outcomes have one buyer, so the count and revenue agree.

Comparing BNN and NBN

In visitor order, B means buy and N means pass.
X counts the buyers among the three visitors.
$Y = 5X$ is total notebook revenue in USD.
Z is 1 if the first visitor buys and 0 if the first visitor passes (unitless).

Predict. Which values agree for these two outcomes?

Outcome	X (buyers)	Y (USD)	Z (unitless)
BNN	1	5	1
NBN	1	5	0

Both outcomes have one buyer, so the count and revenue agree.

The first visitor buys in BNN and passes in NBN, so Z differs.

Probability of exactly one buyer

X counts the buyers among the three visitors.
In visitor order, B means buy (0.4) and N means pass (0.6).

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Predict. Which outcomes give $X = 1$?

Probability of exactly one buyer

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Add the probabilities of BNN, NBN and NNB. These complete outcomes are mutually exclusive.

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$$P(X = 1) = P(\mathrm{BNN}) + P(\mathrm{NBN}) + P(\mathrm{NNB})$$

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Predict. Which outcomes give $X = 1$?
Add the probabilities of BNN, NBN and NNB. These complete outcomes are mutually exclusive.

$$\begin{aligned} P(X = 1) &= P(\mathrm{BNN}) + P(\mathrm{NBN}) + P(\mathrm{NNB}) \\ &= 0.144 + 0.144 + 0.144 = 0.432. \end{aligned}$$

Worksheet reading. The worked school-sale problem completes all three PMFs. The notation reference shows how to write and check the sums.

Probability of 10 USD in revenue

X counts the buyers among the three visitors.
$Y = 5X$ is total notebook revenue in USD.
In visitor order, B means buy (0.4) and N means pass (0.6).

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Predict. Which outcomes give $Y = 10$?

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At 5 USD per notebook, 10 USD means $10 \div 5 = 2$ buyers. The matches are BBN, BNB and NBB.

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$$P(Y = 10) = P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{NBB})$$

Independence lets us multiply the decision probabilities within an outcome. The matching outcomes are mutually exclusive, so we add their probabilities.

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Predict. Which outcomes give $Y = 10$?
At 5 USD per notebook, 10 USD means $10 \div 5 = 2$ buyers. The matches are BBN, BNB and NBB.

$$\begin{aligned} P(Y = 10) &= P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{NBB}) \\ &= 0.096 + 0.096 + 0.096 \end{aligned}$$

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At 5 USD per notebook, 10 USD means $10 \div 5 = 2$ buyers. The matches are BBN, BNB and NBB.

$$\begin{aligned} P(Y = 10) &= P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{NBB}) \\ &= 0.096 + 0.096 + 0.096 = 0.2880. \end{aligned}$$

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Probability for the first visitor

Z is 1 if the first visitor buys and 0 if the first visitor passes (unitless).
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$$\begin{aligned} P(Z = 1) &= P(\mathrm{BNN}) + P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{BBB}) \\ &= 0.144 + 0.096 + 0.096 + 0.064 = 0.4000. \end{aligned}$$

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Independence lets us multiply the decision probabilities within an outcome. The matching outcomes are mutually exclusive, so we add their probabilities.

This equals the original 0.4 probability that the first visitor buys.

Discrete supports and purchase time

Braces list the values in a support.

Variable	Finite support	Units
X counts the buyers among the three visitors.	$\{0, 1, 2, 3\}$	buyers
$Y = 5X$ is total notebook revenue.	$\{0, 5, 10, 15\}$	USD
Z is 1 if the first visitor buys and 0 if the first visitor passes.	$\{0, 1\}$	Unitless

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The **exact time to complete a purchase** in seconds is continuous. It can take values throughout an interval.

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The **exact time to complete a purchase** in seconds is continuous. It can take values throughout an interval.

Purchase time recorded in whole seconds takes discrete values.

The general problem and its three tasks

A situation gives complete outcomes, their probabilities, and information for assigning numbers. The outcomes are mutually exclusive (only one occurs) and exhaustive (they cover every possibility).

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Task 3. Use the PMF to find a requested probability and interpret it in context.

Several complete outcomes may produce the same numerical value.

Task 1: Record outcomes and probabilities

Use one row for *each complete outcome*, even when some outcomes will later have the same numerical value.

Complete outcome	**Probability**
ω_1	q_1
ω_2	q_2
$\vdots$	$\vdots$
ω_n	q_n

Define $q_i = P(\{\omega_i\})$: the probability of the single outcome ω_i.

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$$q_i \geq 0 \quad \text{for every } i, \qquad \sum_{i=1}^{n} q_i = 1.$$

Task 2: Define the variable and map every outcome

Define X in words: what numerical quantity does it measure? State its units, or say that it is unitless.

Keep every outcome and its probability. Add the assigned value:

Complete outcome	**Probability**	$X(\omega_i)$
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$\vdots$	$\vdots$	$\vdots$
ω_n	q_n	$X(\omega_n)$

Each outcome receives exactly one value. Different outcomes can receive the same value.

Task 2: Group values and write the PMF

The outcome table has one row per complete outcome. The PMF has one row per distinct value.

Distinct value x	$p_X(x) = P(X = x)$
x_1	
$\vdots$	
x_m	

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Group: $p_X(x) = \sum_{i:\, X(\omega_i)=x} q_i$. Add all probabilities mapped to x.

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Verify: $p_X(x) \geq 0$ and $\sum_{x \in S_X} p_X(x) = 1$.

Worked model: Buying a carton of fruit

A food stall buys one carton of fruit. Exactly one supplier fills the order. The probabilities for the orchard, cooperative, wholesaler, and grocery are 0.28, 0.19, 0.31, and 0.22, respectively. Their prices for the carton are 14, 18, 14, and 22 USD, in the same order.

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Define C as the purchase cost of one carton, in USD. A complete outcome identifies the supplier.

Predict. Which suppliers produce the same cost? How many rows will the outcome table and the PMF need?

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Define C as the purchase cost of one carton, in USD. A complete outcome identifies the supplier.

Predict. Which suppliers produce the same cost? How many rows will the outcome table and the PMF need?

The orchard and wholesaler both charge 14 USD. Keep four outcomes, but group them into three distinct costs.

Fruit carton: Mapping, PMF and checks

C is the purchase cost of one carton, in USD.

Outcome and mapping table

Supplier	Probability	C
Orchard	0.28	
Cooperative	0.19	
Wholesaler	0.31	
Grocery	0.22	

PMF: group equal costs

c (USD)	$p_C(c) = P(C = c)$

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PMF: group equal costs

c (USD)	$p_C(c) = P(C = c)$
14	$0.28 + 0.31 = 0.59$
18	0.19
22	0.22

Support: $S_C = \{14, 18, 22\}$; values are measured in USD.

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18	0.19
22	0.22

Support: $S_C = \{14, 18, 22\}$; values are measured in USD.
All masses are nonnegative, and

$$0.59 + 0.19 + 0.22 = 1.$$

Task 3: Find and interpret a specific probability

1. Translate the question into an event involving the variable.
2. Read its mass from the PMF, or add the masses for all matching values.
3. Write the probability calculation.
4. Interpret it in context; use a percentage when helpful.

For an exact value: $P(X = a) = p_X(a)$.

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The stall pays exactly 14 USD for the carton in 59% of these orders.

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Submit: outcome table; variable definition and units; mapping; support; PMF with grouping calculations; probability checks; requested probability and contextual interpretation.

Highlight legend

Highlighted item	What it means	Common symbols
Outcome	A possible result of the experiment.	ω (one outcome), Ω (sample space).
Probability	Probability expressions and PMF masses, without units.	$P(\omega)$, $p_X(x)$; e.g. result: 0.4.
First variable	The first variable, its numerical values, and any units.	F, X; values f, x.
Second variable	The second variable, its numerical values, and any units.	E, Y; values e, y.
Third variable	The third variable, its numerical values, and any units.	C, Z; values c, z.
Unknown parameter	A quantity to determine from the information given.	k (illustrative only): this deck uses no unknown parameter.

General model: Mapping, PMF and checks

X is [the numerical quantity measured]; units: [state units or unitless].

Outcome and mapping table

One row per complete outcome.

ω_i	$q_i = P(\{\omega_i\})$	$X(\omega_i)$
ω_1	q_1	
ω_2	q_2	
$\vdots$	$\vdots$	
ω_n	q_n	

PMF: group equal values

One row per distinct value.

x_j	$p_X(x_j) = \sum_{i:\, X(\omega_i)=x_j} q_i$

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Checks: $p_X(x) \geq 0$ for every x, and $\sum_{x \in S_X} p_X(x) = 1$.